Test Edge Cases and Fairness

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

Test Edge Cases and Fairness

> Prepare Test Edge Cases and Fairness: check the target, prerequisites, resources and adaptable teaching schedule.

Teach the learning cycle

Primary target: Choose tests just below, at and above a decision **boundary**

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min

> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Retrieval answer guidance

Test Normal Cases

Without opening the old lesson, explain one key idea from Test Normal Cases. Give an example and a mistake that the idea helps you avoid.

Look for: Plan routine cases with expected results derived from the requirements

Reference example, not the only acceptable wording: R1 B1 because time fits and it is quiet; early match. R2 B2 because B1 is too long, B2 fits and quiet is unrestricted; later match.

R3 no match because every time exceeds 6; exhaustion. R4 no match because B2 fails quiet while other rows exceed 9; both conditions. Record P1 version 1, B version 1 and the manual baseline.

A full pass needs the required **output** and a correct explanation.

Design the Decision Process

Without opening the old lesson, explain one key idea from Design the Decision **Process**. Give an example and a mistake that the idea helps you avoid.

Look for: Design a decision **process** that covers valid, invalid and no-match requests

Reference example, not the only acceptable wording: Clarification follows invalid or missing **input**. A suggestion follows the first candidate passing both compatibility conditions.

No match follows a valid request after all candidates fail. A single failed candidate does not settle whether later candidates fit, so the **process** must advance.

Humans Set Boundaries

Without opening the old lesson, explain one key idea from Humans Set Boundaries. Give an example and a mistake that the idea helps you avoid.

Look for: Distinguish a system’s capability, the correctness of its **output** and permission to take an action

Reference example, not the only acceptable wording: The button establishes a capability. The fact check establishes factual support for A at that time.

The approval permits the specified classroom-board action within its limits. Public posting is outside that permission even though the button exists and A is factually checked.

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

Before testing, define outcomes for clean paper, metal foil, mixed packaging and an unreadable card. The team later changes mixed packaging's expected result to match its sorter. Explain the evaluation problem.

Concept to explain

Unusual and **underrepresented** cases reveal **hidden assumption**s.

Term	Meaning
edge case	unusual input near a system limit
underrepresented	rare or absent in design evidence
subgroup test	check for one relevant condition or user group
hidden assumption	unstated belief revealed by failure

Prepare the class

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

Alternative compact allocation

> This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Boundary testing returns to **inclusive** comparisons and validation using the capstone context. S version 0 is a deliberately faulty comparison example, not the correct P1 implementation.

It can pass the earlier routine set while rejecting exact duration fits. Catalogue variations test later successful paths that the original data cannot reach.

Preserve the distinction between changing data, changing requirements and changing implementation; the variations here change only declared catalogue values.

Readiness answer

They examine only the selected **input**s and conditions. A mistake may affect equality, **invalid input**, another catalogue or a path that none of those cases reaches.

Saved P1 instructions and catalogue B version 1, plus separate labelled copies for versions 2 and 3

Test records with purpose, versions, **input**, expected result, observed result, judgement and trace

Paper cards for the independent equipment selector and a folder retaining every attempted run

End goal for the pictorial case

By the end, I can design **boundary** and coverage tests instead of assuming routine success is enough.

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

TEACHER GUIDE · 2 OF 9

Explain the mechanism

> Model Test Edge Cases and Fairness using the supplied case, labelled mechanism and reasoning. Make the scope of each conclusion explicit.

Follow the information instead of imagining magic inside the machine.

1. Input — information entering

INPUT planned edge and subgroup cases

2. Process — the deciding step

PROCESS run same version and compare error patterns

3. Output — the result

OUTPUT documented gaps, harms, and repair priorities

Key idea

CORE IDEA Unusual and **underrepresented** cases reveal **hidden assumption**s.

How the mechanism works

UNDER THE HOOD Edge testing explores boundaries, rare combinations, missing values, adversarial ambiguity, and subgroup conditions. Fairness analysis asks whether error burdens differ and whether fallback paths remain usable.

Draw a fresh example with all three stages:

Every step connects a claim to observable **evidence**.

The case to investigate

CASE A worn notebook, multilingual label, partly hidden item, and object **fitting** two bins reveal assumptions normal tests miss.

Step 1

STEP 1 — OBSERVE planned edge and subgroup cases

Step 2

STEP 2 — TRACE run same version and compare error patterns

Step 3

STEP 3 — CONCLUDE documented gaps, harms, and repair priorities

Model answer

MODEL REASONING Use the same frozen prototype, log outcomes by condition, and compare error types. Prioritize failures with high burden or no usable fallback, then design a targeted repair without stereotyping users.

A claim to avoid

DO NOT OVERCLAIM **Edge case**s are not excuses to blame users for being unusual; they are **evidence** about design limits.

What new **evidence** could change this answer?

Concrete teaching case

The teaching case

A sorter handles clean paper but fails on crumpled paper and mixed packaging.

The teaching case

Design revealing tests.

Exact explanation to model

Why the result follows

Use separately labelled clean, crumpled and mixed cases with expected **output**s or review actions declared first. Compare conditions.

Explain the added worked example

Explain the added worked example · 1

Routine examples can miss a mistake at an exact limit. This week you will choose revealing cases around those limits and vary the paper catalogue to reach later successful **output**s.

Write expected results from the requirements before operating the prototype. Keep its saved instructions and each test record so a failure remains useful **evidence**.

Explain the added worked example · 2

Catalogue B version 1 is unchanged: B1 Bookmark drawing, 10 minutes, quiet Yes; B2 Story circle, 8 minutes, quiet No; B3 Pattern puzzle, **12 minutes**, quiet Yes; B4 Tower challenge, 15 minutes, quiet No. Search in that order.

These fictional paper values assume materials are available. **Input**s are whole minutes from 0 to 30 **inclusive** and Quiet required Yes or No. No allows quiet or nonquiet activities.

Explain the added worked example · 3

Time fits when required minutes are no more than available minutes, including equality. The quiet condition must also hold. Select the first **fitting** row with a correct reason, or report no match after every row fails.

Invalid or missing **input** requires clarification before searching. P1 validates, resets to the first row, checks both conditions, advances after failure and stops at the first fit or catalogue exhaustion.

No **output** books or publishes anything.

Explain the added worked example · 4

For quiet requests, 10 minutes is the first time a catalogue activity fits. Test 9, 10 and 11 with Yes so only the time changes.

A faulty comparison S version 0 uses strictly less than instead of no more than for time, while keeping the other P1 rules. It rejects B1 at exactly 10.

With quiet required, every later row also fails, so it incorrectly reports no match.

Explain the added worked example · 5

S can pass all four routine requests from Week 33: 11 Yes, 9 No, 6 No and 9 Yes. None has a **fitting** activity whose duration equals the request. The 10 Yes case reveals a gap those passes did not close.

Another useful **boundary** is 8 minutes with No: equality should allow B2, while 7 No has no match. Keep both quiet values in your wider plan because No does not mean nonquiet only.

Key terms

Term	Meaning
Boundary	A value where an input changes category or a condition changes truth
Inclusive	Including the stated endpoint or equal value
Invalid input	A value that does not meet the agreed input rules
Test variation	A documented change in the conditions used for a test
Coverage gap	A relevant condition or behaviour the current tests have not examined

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Request using catalogue B	Expected response	Reason
0 minutes Yes	No match	Zero is valid; every activity needs more time
30 minutes Yes	B1 with a matching reason	30 is valid; B1 is the first fit
31 minutes Yes	Clarify minutes; no search	31 is outside the accepted range
Blank minutes with Yes	Clarify minutes; no search	The time field is missing
10.5 minutes Yes	Clarify minutes; no search	The accepted time must be a whole number

Request using catalogue B	Expected response	Reason
10 minutes with blank quiet	Clarify quiet; no search	Choose Yes or No before searching

Model the reasoning aloud · 6

The accepted range and activity durations are different kinds of **boundary**. Zero belongs to the valid range even though no activity fits. Thirty also belongs to it.

A negative time, such as minus one minute, is below zero and invalid. Do not change a blank to zero, round 10.5 to 10 or turn a blank quiet choice into No. Ask for the field to be corrected and start a new validated run.

Model the reasoning aloud · 7

A test record should say whether a search is expected to begin. Clarification should name the field needing correction; it should not claim the catalogue has no suitable activity.

Test an unrecognised quiet value such as Maybe as well as a missing value. These check different forms of unusable **input**.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Test Edge Cases and Fairness

A fictional craft-book chooser accepts requests for whole available minutes from 0 to 20.

The facts we will use

Supplied facts

Booklet A takes 6 minutes; B takes 12. Return first duration at most available time.

Supplied facts

Check 5,6,7; also 0,20,21 and blank. Same catalogue applies to every user.

AI ACADEMY / AI-1/34

Test Edge Cases and Fairness

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Near a boundary

At 5: none. At 6 and 7: A.

2. Valid extremes

0: none. 20: A, because first-fit order remains fixed.



3. Invalid inputs

21 and blank require clarification, not ordinary search.



4. Coverage record

Record each user condition tested; do not invent unrun outcomes.

CHANGE ONE THING

Faulty code uses duration strictly below available time.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Near a boundary	At 5: none. At 6 and 7: A.
2. Valid extremes	0: none. 20: A, because first-fit order remains fixed.
3. Invalid inputs	21 and blank require clarification, not ordinary search.
4. Coverage record	Record each user condition tested; do not invent unrun outcomes.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Adjacent values below, at and above 6 expose an accidental strict inequality. Zero is valid even though no activity fits.

The top valid **input** does not require selecting the longest activity: catalogue order still controls selection. Coverage also asks whose conditions were tested, which cannot be inferred from these numbers alone.

What this example does not establish

Keep the stated limit

Equal **written rule**s do not prove equally good outcomes for every group; actual condition-specific **evidence** is still needed.

Change one thing in the pictured case

Change one thing

Change: Faulty code uses duration strictly below available time.

Predicted result

Prediction: Request 6 now wrongly returns none.

Why the result follows

Reason: A's duration equals the available time and should be eligible under at most.

Ask while showing the picture

Ask while showing the picture · 8

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

Demonstrate and guide practice

> Demonstrate Test Edge Cases and Fairness, reduce support deliberately and check individual reasoning before discussing answers.

Board sequence

1. Put the facts on the board

Write the given facts: A sorter handles clean paper but fails on crumpled paper and mixed packaging.

2. State the decision to explain

Write the question to explain: Design revealing tests.

3. Reveal the worked reasoning

Reveal after **prediction**s: Use separately labelled clean, crumpled and mixed cases with expected **output**s or review actions declared first. Compare conditions.

Hold back the independent task

Keep for an independent attempt: A sorter works on clear photos. Test a blurred pencil, a dim book and an unfamiliar lunch box. What makes these useful tests?

Classroom dialogue

Ask the class

Ask: Which facts are given, and which would we have to assume? Learners mark relevant

details before discussing answers.

Language to model

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

Model these guided questions

For B version 3 and 15 No, B1 20, B2 18 and B3 17 fail time. B4 15 fits exactly and quiet is unrestricted, so select B4 and stop. For 15 Yes, B4 fails quiet and all earlier rows fail time, so no match after all four.

Record B version 3, the saved instruction version, exact request, expected and observed **output** with reasons, and trace.

Exit: inspect strictly less than versus no more than; the specification includes equality, so expected 10 Yes remains B1. Another equality case is 8 No, expected B2.

Invalid 31 Yes requires minutes clarification before searching.

Case-specific teaching sequence

Stage	Teacher action
Session 1 start to 7 minutes	Retrieve why routine-case success can leave coverage gaps.
7 to 20 minutes	Use the diagram and W1 to compare below, at and above ten minutes.
20 to 33 minutes	Complete W2; distinguish accepted range endpoints from activity duration limits.
33 to 45 minutes	Prepare B version 2 and begin W3 expected results and partner traces.

Stage	Teacher action
Session 2 start to 8 minutes	Finish W3 and verify both methods used the same catalogue copy.
8 to 23 minutes	Trace guided version 3 requests and discuss purpose and version labels.
23 to 38 minutes	Collect W4 independently before support; inspect both equality cases and validation.
38 to 45 minutes	Complete W5 and the exit; save the expanded development set for repair checks.

Reduce help deliberately

Completion practice

Completion task: Classify these requests before searching: 0 Yes, 30 Yes, 31 Yes, blank minutes with Yes, 10.5 Yes, and 10 Maybe. Give each expected **output** or clarification and explain whether any candidate should be checked.

Why must zero not be treated like a blank?

Reasoning to listen for

Teacher check: 0 Yes is valid and checks all rows before no match. 30 Yes is valid and selects B1 at the first row. 31, blank minutes and 10.5 require minutes clarification; 10 Maybe requires a Yes or No quiet choice.

Invalid cases check zero candidates. Zero is a supplied whole number within range; blank is missing information.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for Testing boundaries and varied conditions. Proposed explanation: “At the limit means outside the rule”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> **Expected repair**
>
> Expected correction: An **inclusive** rule accepts its endpoints and equality.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial practice question

> Which single **boundary** run most directly detects the strict-comparison bug, and why?

Reasoned teacher answer

6: intended A, faulty none. **Input**s 5 and 7 alone would miss this exact equality defect.

Changed-case reasoning: Request 6 now wrongly returns none. A’s duration equals the available time and should be eligible under at most.

Collect each learner’s explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Run the weekly evidence lab

> Run the Test Edge Cases and Fairness investigation and use the first unsupported step to choose a teaching response.

Predict first, test honestly, and explain what the **evidence** means.

1. Predict

State your **prediction** before the result is known.

2. Investigate

Complete the task: Add missing, ambiguous, and **subgroup test** cards.

3. Compare cases

Keep a normal case and an **edge case** clearly separated.

4. Record honestly

Record what happened—even when it does not match your **prediction**.

5. Explain and plan the next test

Explain the result, one limitation, and the next useful test.

Use a fresh case after support

INDEPENDENCE After support is removed, repeat the proof with a fresh case.

Make the lab runnable

Use the concrete case as the shared normal case and the independent question as a second case. Each pair creates a **boundary** or missing-information case and predicts its result before testing.

Keep the expected result separate from an observed mistake.

Record	What to capture
Input or evidence	Exact data, instruction or claim
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome
Next test	One justified change and a new prediction

Diagnose and reteach

Find the unsupported claim

CLAIM TO REPAIR Repeating only easy **input**s tests the system's limits.

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

At the limit means outside the rule: An **inclusive** rule accepts its endpoints and equality.

No match and invalid mean the same thing: No match follows valid **input**; invalid fields need clarification before search.

A catalogue change needs the same expected answer: Recompute expectations from the declared new data and unchanged requirements.

More routine repetitions close every gap: Choose cases that exercise a missing condition or path.

Match the support to the first error

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a **boundary** or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

Assess and explain the answers

> Use the Test Edge Cases and Fairness questions, teacher keys and criteria after the relevant attempts. Previously discussed cases are practice; protect the separately selected independent task.

Worked case answer

> **Teacher key · Worked case answer**
>
> Use separately labelled clean, crumpled and mixed cases with expected **output**s or review actions declared first. Compare conditions.

Independent question

> **Independent question · withhold the key**
>
> A sorter works on clear photos. Test a blurred pencil, a dim book and an unfamiliar lunch box. What makes these useful tests?

Expected independent answer

> **Teacher key · Expected independent answer**
>
> They probe unclear **input**, changed lighting and an unknown category. Record expected behaviour before each test, including review when **evidence** is insufficient.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: “Repeating only easy **input**s tests the system's limits.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

Teacher key · Original transfer answer

A reasonable policy is P, M, review, review. Changing expected labels just to match **prediction**s makes the test circular.

If the policy itself was wrong, document a justified correction and reevaluate all affected cases, preserving the original record.

Explanations for the additional workbook examples

W1 Plan the equality tests

Workbook question · W1

For catalogue B version 1, write expected **output**s and reasons for 9 Yes, 10 Yes and 11 Yes. Explain why keeping quiet and catalogue fixed helps examine the time **boundary**.

Predict what the faulty strictly-less-than S comparison does for 10 Yes.

Teacher reasoning · W1

9 Yes has no match: B1 and B3 are too long and B2 is not quiet; B4 also fails. 10 Yes and 11 Yes select B1, which is quiet and fits at equality or with time left. Fixing the other conditions isolates the time comparison.

S rejects equality at B1 and cannot find a later fit for 10 Yes, so incorrectly returns no match.

W2 Check the validation boundary

Workbook question · W2

Classify these requests before searching: 0 Yes, 30 Yes, 31 Yes, blank minutes with Yes, 10.5 Yes, and 10 Maybe. Give each expected **output** or clarification and explain whether any candidate should be checked.

Why must zero not be treated like a blank?

Teacher reasoning · W2

0 Yes is valid and checks all rows before no match. 30 Yes is valid and selects B1 at the first row. 31, blank minutes and 10.5 require minutes clarification; 10 Maybe requires a Yes or No quiet choice.

Invalid cases check zero candidates. Zero is a supplied whole number within range; blank is missing information.

W3 Run a labelled catalogue variation

Workbook question · W3

Copy B version 1 and change only B1's duration to 20, labelling it B version 2. Write the expected result for 12 Yes, then run your saved P1 and the manual baseline on this copy. Record the actual traces separately.

Explain why an answer checked against B version 1 would not be a fair judgement of this run.

Teacher reasoning · W3

Expected B3: B1 takes too long, B2 fails quiet, B3 fits 12 exactly and is quiet. Correct P1 and baseline agree, but retain actual disagreements for investigation.

Under version 1 the expected first fit is B1, so using that expectation with changed data would mix conditions. Save catalogue and instruction versions with both traces.

W4 Find an independent capacity boundary

Workbook question · W4

A new equipment selector lists E1 first, capacity 4; E2 second, capacity 8. Valid group size is a whole number from 1 to 10.

Choose the first item whose capacity is at least the group size, or no match; **invalid input** needs clarification. A faulty version uses capacity strictly greater than group size.

Independently give correct and faulty **output**s for groups 3, 4 and 5, then for group 8. Explain the defect and give the correct response to group 0. No booking occurs.

Teacher reasoning · W4

Correct **output**s for 3, 4, 5 and 8 are E1, E1, E2 and E2. Faulty **output**s are E1, E2, E2 and no match. Equality is wrongly excluded at capacities 4 and 8. Use capacity at least group size, including equality.

Group 0 is outside the accepted 1 to 10 range and needs clarification before any search. This range differs from the activity time range.

W5 Audit what changed in the test

Workbook question · W5

A team runs 15 No on B version 3, gets B4, but records “B version 1” and calls B4 correct. Explain the documentation error and the expected result under each catalogue.

State what must stay the same in a fair prototype-versus-baseline comparison and why a test not yet run cannot be counted as a pass.

Teacher reasoning · W5

For version 3, B1 20, B2 18 and B3 17 are too long; B4 15 fits and quiet is unrestricted, so B4 is correct. Under version 1, earlier B1 fits and must be selected.

Correct the catalogue label while preserving what was actually used and observed. Both methods need the same request, catalogue and pass criterion. An unrun case has no observed **evidence**, so cannot be counted as passing.

Assess independent transfer

Independent question · withhold the key

Selected case: A new equipment selector lists E1 first, capacity 4; E2 second, capacity 8. Valid group size is a whole number from 1 to 10.

Choose the first item whose capacity is at least the group size, or no match; **invalid input** needs clarification. A faulty version uses capacity strictly greater than group size.

Independently give correct and faulty **output**s for groups 3, 4 and 5, then for group 8. Explain the defect and give the correct response to group 0. No booking occurs.

Selected case answer

Teacher key · Selected case answer

Correct **output**s for 3, 4, 5 and 8 are E1, E1, E2 and E2. Faulty **output**s are E1, E2, E2 and no match. Equality is wrongly excluded at capacities 4 and 8. Use capacity at least group size, including equality.

Group 0 is outside the accepted 1 to 10 range and needs clarification before any search. This range differs from the activity time range.

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.

Dimension	Evidence required
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

> **Teacher key · Selected case answer**
>
> Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.
>
> Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

Connect and extend learning

> Connect Test Edge Cases and Fairness to Helpful Classroom Sorter, adapt support and plan a later changed-case check.

Helpful Classroom Sorter

Expand the sorter test suite with edge, missing-data, and fairness cases plus expected safe behavior.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

Support and stretch

Use physical cards, spoken explanations and a three-stage drawing. Read aloud without giving the answer. Stretch: compare two **mechanism**s with the same visible **output** and explain what **evidence** distinguishes them.

Exit reflection

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Bridge to the next week

Next: Week 35: Improve One Cause. Retrieve today's **evidence** and identify what carries forward and what is new.

Support and fresh reassessment

Draw a number line around each **boundary** and read at least or no more than aloud. Keep validation and candidate checks on separate cards. Provide version-label headings without filling expected results.

Extension: contrast a catalogue reorder with a duration change and explain how the first-match requirement remains unchanged in both.

W4 should distinguish correct E1, E1, E2, E2 from faulty E1, E2, E2, no match for 3, 4, 5, 8. Require an **inclusive**-capacity explanation and clarification for zero, not the activity task's valid-zero rule.

After support, use a new catalogue F1 capacity 5 then F2 capacity 9 with the same 1 to 10 **input** range. For groups 5 and 9, correct F1 and F2; strict comparison gives F2 and no match.

Require an independent explanation of both changes.

Week 35 repairs a documented defect, reruns revealing cases and checks earlier successful cases for regressions before fixing a version for final evaluation.

Plan the delayed return

Suggested return points in this level: ai-1/35. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

Add sorter **boundary** cases, missing fields and relevant user conditions to the test plan before making fairness claims.

Prepare, teach and assess

> Use this preparation summary alongside the detailed Test Edge Cases and Fairness lesson; it

> is a checklist rather than a replacement for the full sequence.

Teaching target

Teaching target: Choose tests just below, at and above a decision **boundary**

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. No paid AI account is required for the paper reasoning activities.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

Before class, solve the supported practice: Classify these requests before searching: 0 Yes, 30 Yes, 31 Yes, blank minutes with Yes, 10.5 Yes, and 10 Maybe.

Give each expected **output** or clarification and explain whether any candidate should be checked. Why must zero not be treated like a blank?

Supported practice answer

Reasoned practice answer: 0 Yes is valid and checks all rows before no match. 30 Yes is valid and selects B1 at the first row. 31, blank minutes and 10.5 require minutes clarification; 10 Maybe requires a Yes or No quiet choice.

Invalid cases check zero candidates. Zero is a supplied whole number within range; blank is missing information.

Check every learner before discussion: Use the worked case for Testing boundaries and varied conditions. Proposed explanation: “At the limit means outside the rule”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Repair the misconception

Misconception repair: An **inclusive** rule accepts its endpoints and equality.

Support without giving away the answer

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extend the reasoning

Extension: Faulty code uses duration strictly below available time. Require a **prediction** and a reason before comparing. Expected result: Request 6 now wrongly returns none.

Reason: A’s duration equals the available time and should be eligible under at most.

Protect the independent attempt

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint

Project checkpoint: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with **evidence**.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

Use feedback, labs and repair paths

> Use feedback, earlier review and portfolios purposefully. Apply the coding routine only when this lesson has an executable lab.

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

For later coding lessons

> This week has no code block. Use this routine when a later lesson includes an executable lab.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-**boundary** and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

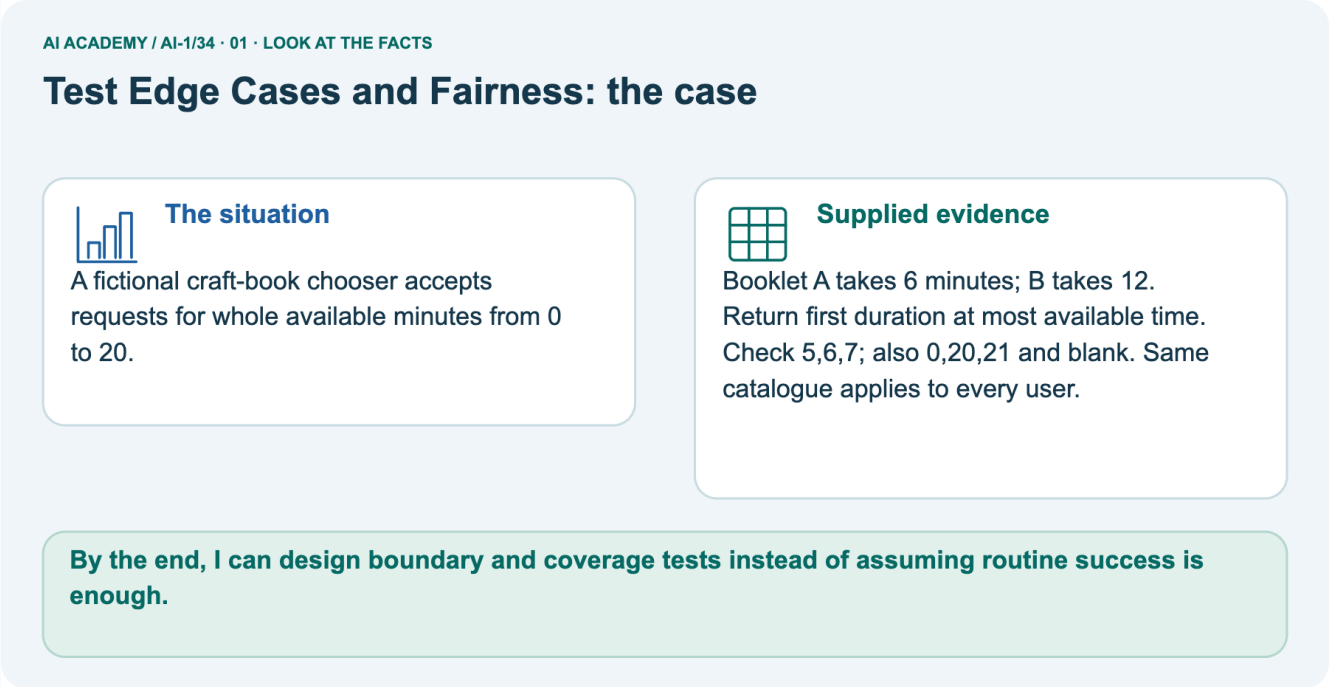
Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

Visual concept journey

> Show one Test Edge Cases and Fairness graphic at a time, request a prediction and discuss its trace before the independent assessment.

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Test Edge Cases and Fairness: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional craft-book chooser accepts requests for whole available minutes from 0 to 20. Supplied facts: Booklet A takes 6 minutes; B takes 12. Return first duration at most available time.

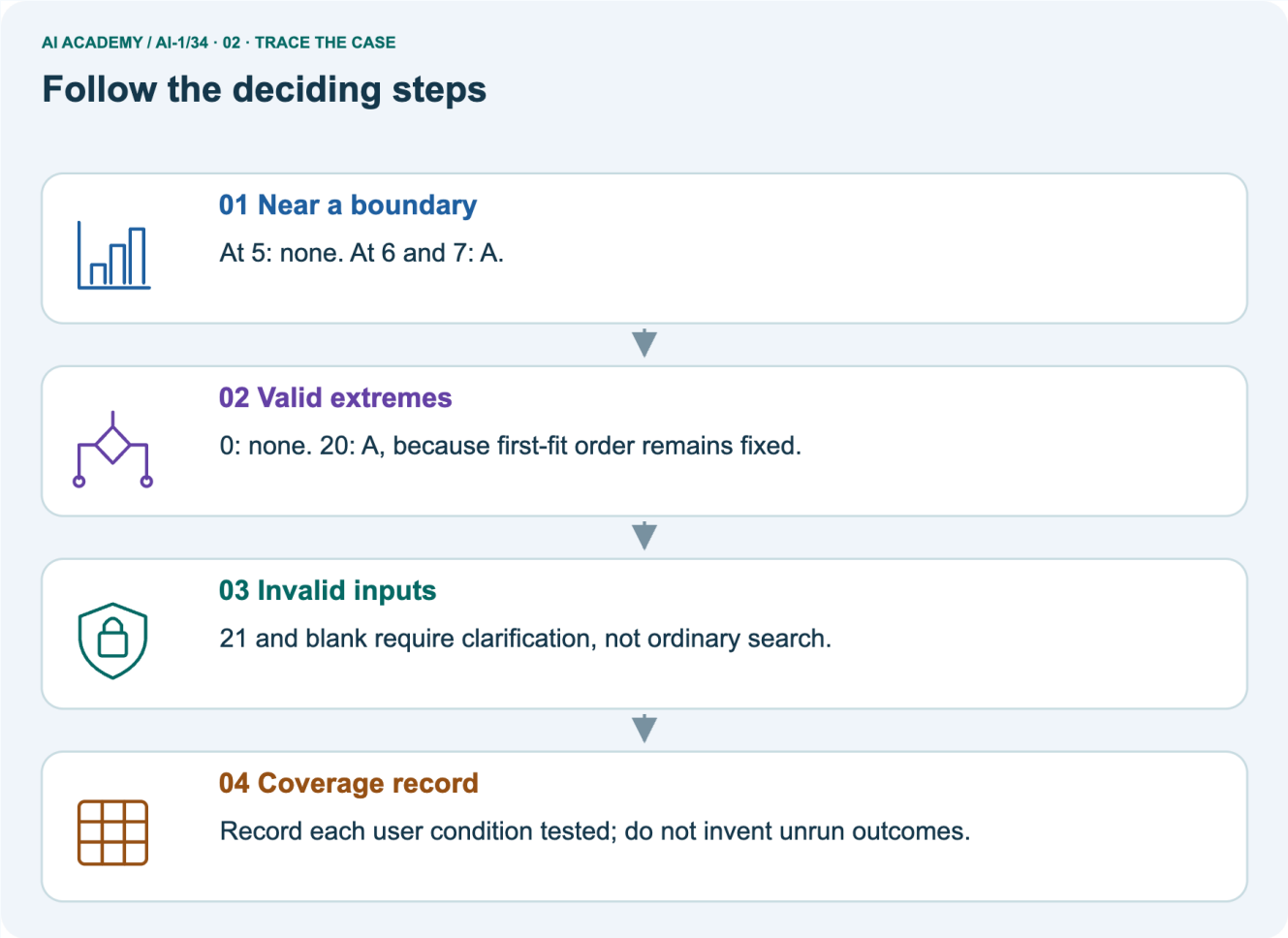
Check 5,6,7; also 0,20,21 and blank. Same catalogue applies to every user.

Point to the given fact that matters most. Explain why it matters.

Discuss while showing the picture

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Near a **boundary**: At 5: none.

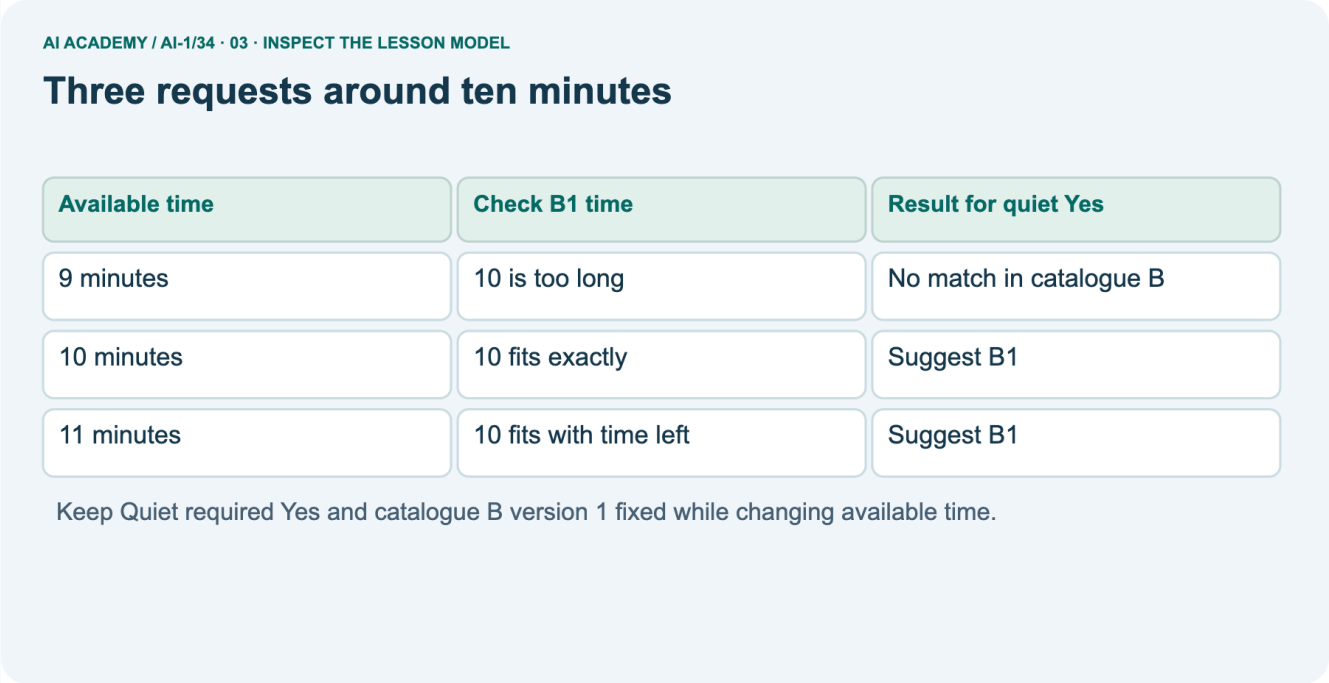
At 6 and 7: A. → Valid extremes: 0: none. 20: A, because first-fit order remains fixed. → **Invalid input**s: 21 and blank require clarification, not ordinary search. → Coverage record: Record each user condition tested; do not invent unrun outcomes.

> Point to each numbered stage and explain which recorded **input** it uses.

Discuss while showing the picture

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Three requests around ten minutes



Follow the labelled picture with the matching explanation. Open the vector diagram

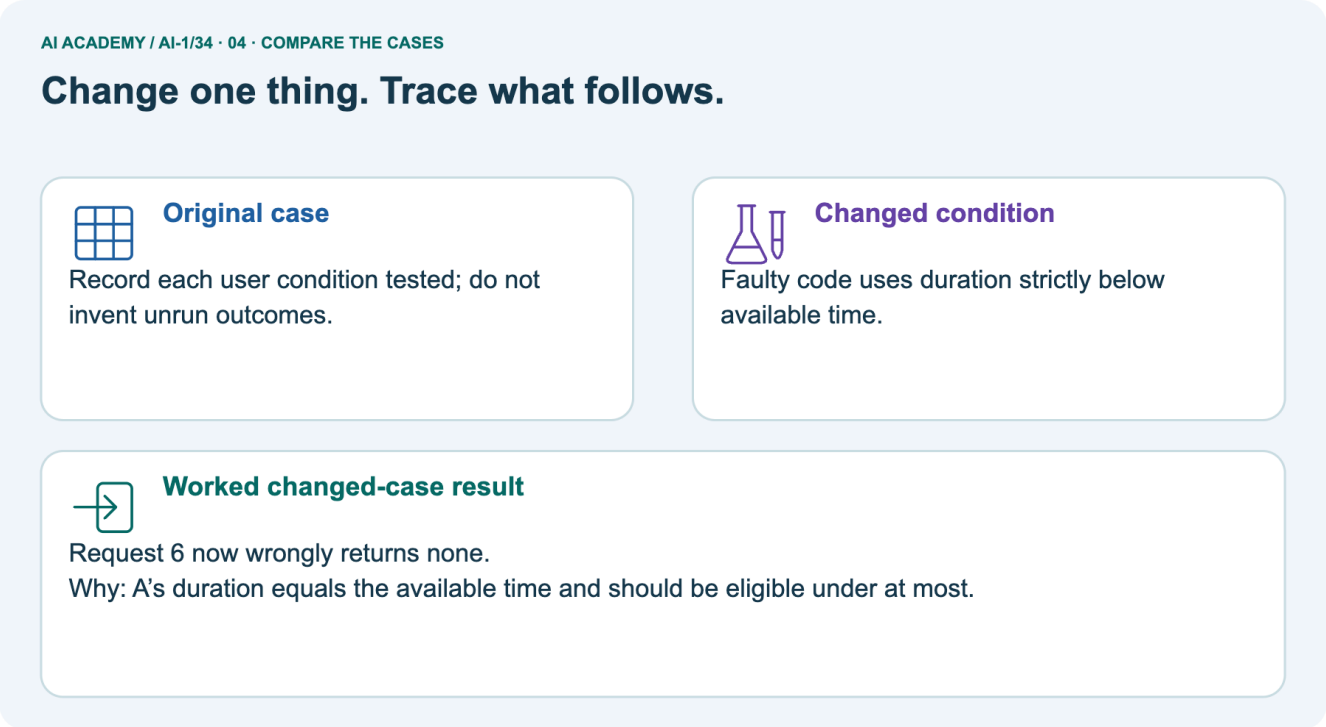
Three requests around ten minutes. 9 minutes | 10 is too long | No match in catalogue B. 10 minutes | 10 fits exactly | Suggest B1. 11 minutes | 10 fits with time left | Suggest B1

Read the labels and values. Explain the relationship the model shows.

Discuss while showing the picture

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Record each user condition tested; do not invent unrun outcomes. Changed condition: Faulty code uses duration strictly below available time. Expected result: Request 6 now wrongly returns none.

Reason: A's duration equals the available time and should be eligible under at most.

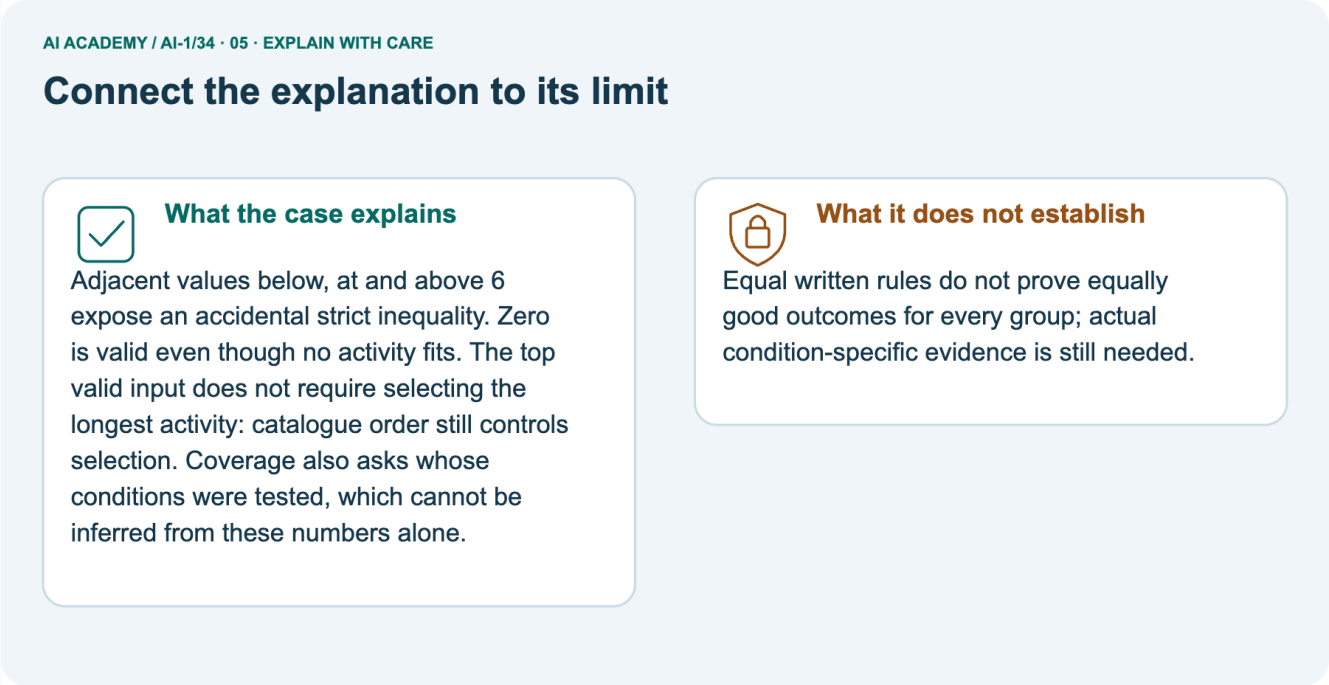
Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Discuss while showing the picture

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: A's duration equals the available time and should be eligible under at most.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Adjacent values below, at and above 6 expose an accidental strict inequality. Zero is valid even though no activity fits.

The top valid **input** does not require selecting the longest activity: catalogue order still controls selection. Coverage also asks whose conditions were tested, which cannot be inferred from these numbers alone.

Limit: Equal **written rule**s do not prove equally good outcomes for every group; actual condition-specific **evidence** is still needed.

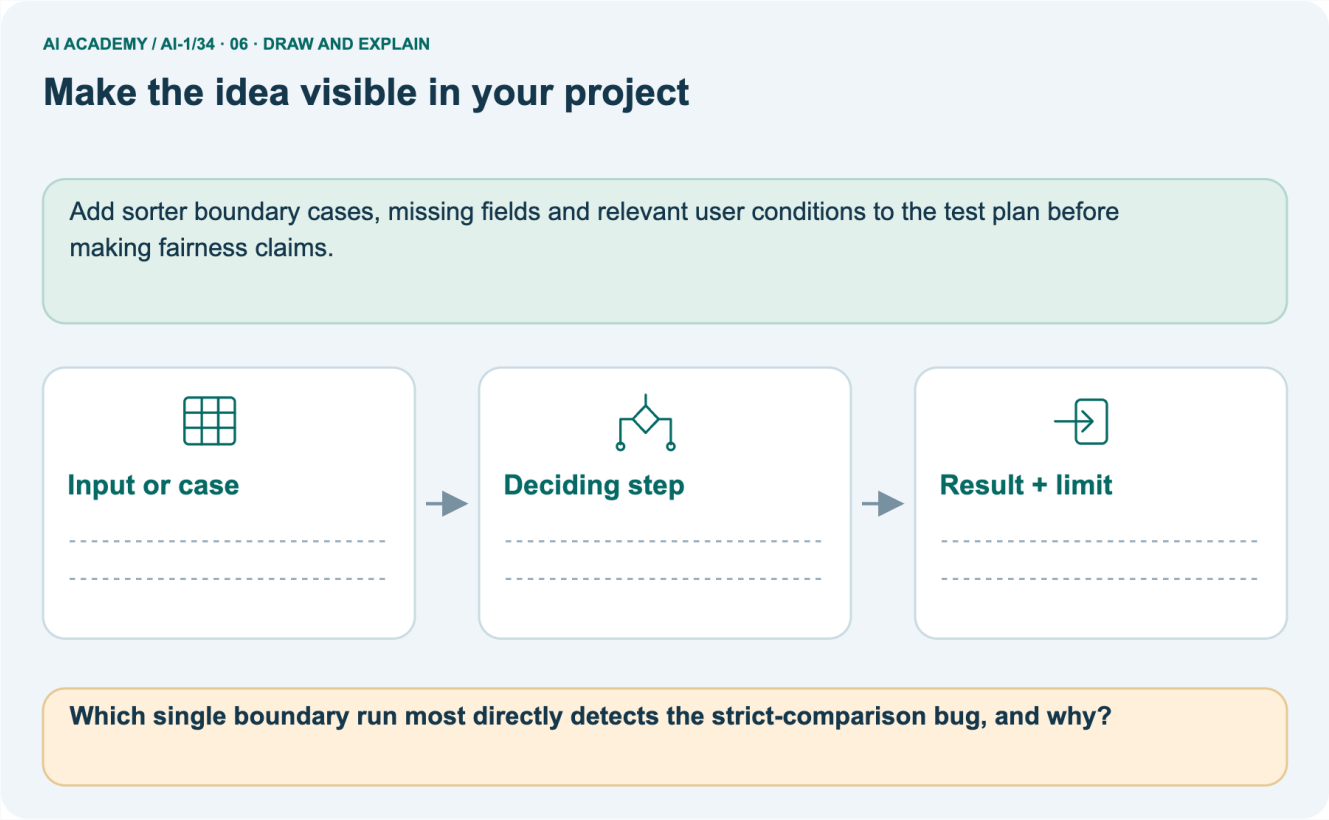
Explain the case, then point to a limit on the claim.

Discuss while showing the picture

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: Equal **written rule**s do not prove equally good outcomes for every group; actual condition-specific **evidence** is still needed.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Add sorter **boundary** cases, missing fields and relevant user conditions to the test plan before making fairness claims. Draw your **input**, deciding step, result and limit.

Supported question: Which single **boundary** run most directly detects the strict-comparison bug, and why?

Draw your project trace and justify the deciding step using **evidence**.

Discuss while showing the picture

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: 6: intended A, faulty none. **Input**s 5 and 7 alone would miss this exact equality defect.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.